

Sushant Nagil

DevOps & Cloud Infrastructure Engineer

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Summary

DevOps and Cloud Engineer, 3 years running multi-account AWS and EKS for state-utility smart-metering rollouts in India. The platform now serves 1.5 million meters, with a head-end sized for 100,000+ concurrent connections. Day to day that is Kubernetes autoscaling (HPA, VPA, KEDA, Karpenter), landing-zone and IAM governance, on-call, and FinOps work that has taken roughly \$45K/yr out of the bill. CKA, AWS Solutions Architect Associate, AWS Data Engineer Associate, HashiCorp Terraform Associate.

Experience

Polaris Smart Metering

May 2025 – Present

DevOps Engineer

Jaipur, India

- Run layered autoscaling across four production EKS clusters carrying data for 1.5 million meters (HPA, Goldilocks-guided VPA, KEDA on Kafka lag, Karpenter), and moved every workload off managed node groups.
- Own FinOps for a 24-account AWS estate, with about \$45K/yr in recurring savings from right-sizing, a zombie node group nobody had noticed, CloudWatch-agent cleanup across 6 accounts, and 100% Aurora RI coverage.
- One of three engineers on the infrastructure team; own the landing zone (OU layout, SCP guardrails, cost-allocation tagging) and review every change that touches shared production accounts.
- Set ECR retention policies across 500+ repositories, then ran a phased cleanup of 167 stale pipelines, build projects, and registries, backing up each batch before deleting it.
- Built the VPC and hybrid networking (per-tier private subnets, site-to-site VPN into on-prem DC gear); compared Transit Gateway against VPC peering and stayed on peering at our account count.
- Set up Postgres DR on EC2 with pgBackRest physical and logical backups instead of streaming replicas, which held RTO and RPO steady without doubling compute. Also locked the S3 IAM path so backups cannot be deleted.
- Carry infra on-call and run postmortems across HES and MDMS. Recent root causes: a Redis split-brain behind a headless service, under-replicated Kafka partitions, a WAF rate-limit that looked like a CORS bug.
- Wired cross-account alerting from CloudWatch through SNS and Lambda into Telegram and Slack, so alarms from every account land in one place.

Polaris Smart Metering

Nov 2023 – Apr 2025

AWS Cloud Engineer

Jaipur, India

- Designed the IAM and AWS SSO model across the accounts, with least-privilege roles and per-team permission sets, and moved the team off standalone IAM users.
- Ran the AWS MAP migration assessment and wave plan, then moved Redis from EC2 to ElastiCache and Kafka from EC2 to MSK, which removed two of our noisiest on-call sources.
- Stood up an EMQX MQTT cluster sized for 100,000+ concurrent meter connections feeding the head-end system.
- Cut non-production AWS spend by ~40% by scheduling dev and stage shutdowns with Lambda and EventBridge.
- Built CodePipeline CI/CD for Lambda jobs and scheduled tasks, which took the manual steps out of dev and stage deploys.
- Tuned self-managed Postgres on EC2 (pgtune, PgBouncer pooling) for the metering workload, ran RabbitMQ clusters, and wrote the vacuum and restore runbooks the team still uses.
- Built the observability stack (Prometheus, Grafana, Loki, Fluent Bit, OpenTelemetry alongside CloudWatch) with SNS alerting to Slack and email for first-line on-call.

Polaris Smart Metering
Cloud Engineering Intern

Aug 2023 – Oct 2023
Jaipur, India

- Mapped the existing AWS footprint (accounts, VPCs, EC2 fleet, security groups) into a reference doc the team used during the later MAP assessment.
- Set up the first CloudWatch alarms and dashboards for the EC2 fleet, and tagged untagged resources so spend could be split by environment.
- Shadowed on-call to learn the metering stack end to end, then converted to a full-time cloud engineer.

Skills

- **Kubernetes:** Kubernetes (CKA), Amazon EKS, Karpenter, KEDA, HPA/VPA, Goldilocks, Helm, ArgoCD, Docker
- **AWS:** EC2, Lambda, EventBridge, VPC/VPN, ELB/NLB, Route 53, S3, IAM & SSO, Organizations & SCPs, ECR, CloudWatch, RDS/Aurora, MSK, ElastiCache, WAF
- **IaC:** Terraform (modules, remote state), CodePipeline, CodeBuild, Git/GitHub
- **Reliability & Observability:** On-call, incident response, postmortems, SLO/SLA, Prometheus, Grafana, Loki, Fluent Bit, OpenTelemetry, Zabbix
- **Data & Messaging:** PostgreSQL, pgBackRest, PgBouncer, Redis, Apache Kafka, RabbitMQ, EMQX MQTT
- **FinOps:** Rightsizing, Savings Plans & Reserved Instances, ECR lifecycle, multi-account cost allocation

Certifications

- **AWS Certified Data Engineer, Associate:** May 2026
- **Certified Kubernetes Administrator (CKA):** The Linux Foundation, Dec 2025
- **HashiCorp Certified: Terraform Associate:** Sep 2025
- **AWS Certified Solutions Architect, Associate:** Jul 2025

Education

B.K. Birla Institute of Engineering & Technology
B.Tech, Computer Science

2020 – 2024
Pilani, India

Volunteer Experience

National Cadet Corps (NCC)
Under Officer, C Certificate holder

Rajasthan, India